# Simulator Laboratory

**Thermodynamics and Thermal Power Plant Simulator**

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# Preface

Simulator lab exercises (SIM LABS) serve as instructional tools within Thermodynamics and Thermal Power Plant Simulator (TPPS) courses. Students in Power Engineering frequently encounter difficulty with theoretical concepts that benefit from additional pedagogical support, while instructors must navigate limited instructional time when integrating both theoretical and practical components. SIM LABS help bridge this gap by strengthening students' understanding of core ThermoFluids principles and supporting the application of these principles in realistic operational contexts. By allowing students to visualize processes, explore system behaviour, and make cause-and-effect connections, SIM LABS enhance conceptual comprehension and better prepare learners for hands-on work in a physical plant using real equipment. In addition, the SIM LABS reinforce safe operational practices and environmental awareness within steam plant systems.

# Chapter 1

# Boiler Efficiency

**Learning Objectives**

Operate the plant in the following modes and compute the boiler thermal efficiency:

- 230 MW, burning HFO
- 80% load, burning biofuel
- 80% load, burning coal
- 80% load, burning coal with soot buildup

## 1.1 Theory

First, we define efficiency. There are many ways to present it; for example, efficiency is the ability to perform a task without wasting energy, effort, or time. Mathematically, it is the ratio of useful output to thermal energy input.

Laypersons often use the terms *efficiency* and *effectiveness* interchangeably; however, efficiency relates to minimizing waste, whereas effectiveness relates to maximizing output (more on this in Chapter 5).

Boiler efficiency is sometimes defined as combustion efficiency, which is determined by the burner's ability to completely burn fuel, accounting for unburnt fuel and excess air in the exhaust. Thermal efficiency, on the other hand, indicates the heat exchanger's (i.e., the boiler's) ability to transfer heat from the combustion process to the water or steam.

In general, the maximum attainable boiler efficiency depends on factors such as the fuel-burning method, furnace design, and heat transfer surfaces. In addition, fuel type, boiler load, and operating practices influence efficiency. In this lab, we focus on fuel type, boiler load, and best practices.

## 1.2 Boiler Thermal Efficiency

Next, we express the boiler thermal efficiency as follows:

$$\eta_{\text{boiler}} = \frac{\text{Energy to steam}}{\text{Energy from fuel}}$$

Here, the energy to steam is the heat transfer required to generate steam. Let:

- $h_2$ = specific enthalpy of steam [kJ/kg]
- $h_1$ = specific enthalpy of feedwater [kJ/kg]

Because steam is generated at constant pressure, the heat required to produce 1 kg of steam is:

$$\text{Energy to steam} = (h_2 - h_1) \quad [\text{kJ}]$$

Energy from fuel is calculated from the mass of fuel used and its calorific value. For coal, this is the heating value measured in a bomb calorimeter, corresponding to the internal energy of combustion. Let:

- $m_f$ = mass of fuel burned over a given time
- $m_s$ = mass of steam generated over the same time
- $HV$ = heating value of the fuel [kJ/kg]

$$\text{Energy from fuel} = m_f \times HV \quad [\text{kJ}]$$

Thus, the boiler thermal efficiency is:

$$\eta_{\text{boiler}} = \frac{m_s(h_2 - h_1)}{m_f HV} \times 100\%$$

Lab Instructions

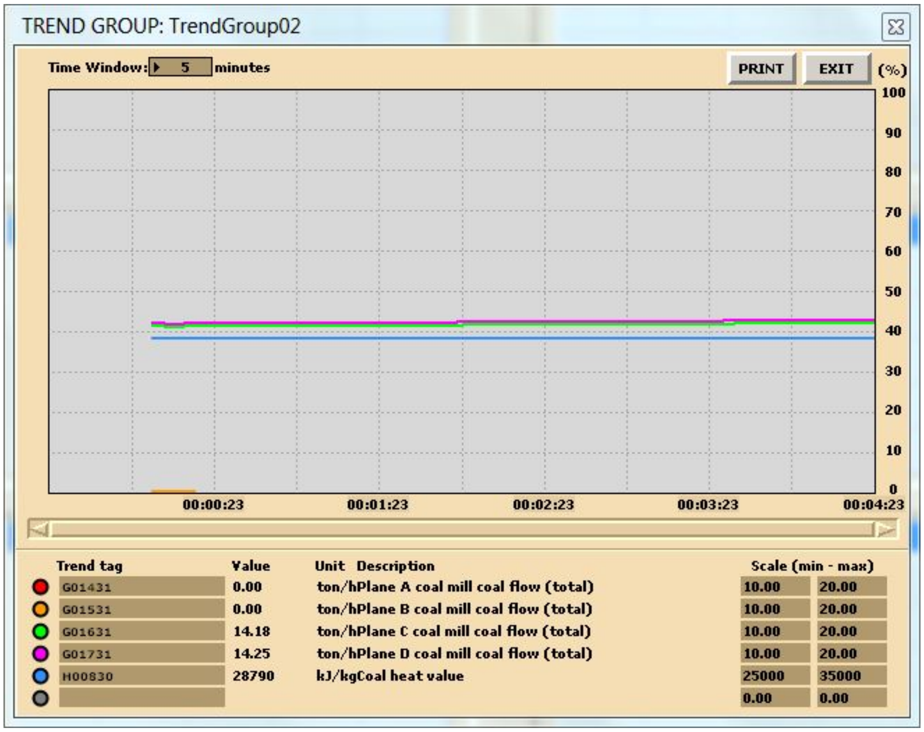
You will run four different initial conditions in this lab:

- **I10:** 230 MW, burning HFO
- **I15:** 80% load, burning biofuel
- **I14:** 80% load, burning coal
- **I14:** 80% load, burning coal with soot (use MD250 to configure soot variables)

For each condition, collect the relevant data to compute the boiler thermal efficiency.

## 1.3 Hints & Tips

For data collection, use trends as shown in the lab manual (see *Trends sample*).

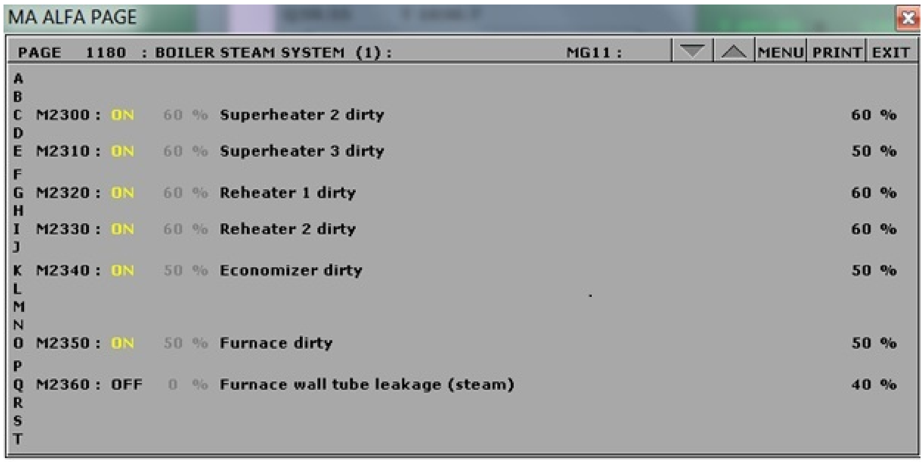



In addition to pressure, temperature, and flow measurements, log the following tags in your trends:

| Tag | Description |
|---|---|
| H00810 | HFO heating value |
| H00870 | Pellet heating value (biofuel) |
| H00830 | Coal heating value |

To calculate enthalpy values, you may use an app or an online tool such as the Superheated Steam Table.

For the coal operation with soot buildup, use MD250 and set malfunctions as shown in the lab manual (see *MD250 malfunction settings*).




**!** Deliverables

Your lab report must include:

- **Trend plots:** All plots for each of the four conditions
- **Computation:** Calculate boiler thermal efficiency for the specified conditions
- **Conclusion:** A summary (max. 500 words; use a text box if using Excel) comparing results and suggesting areas for further study

## 1.4 Further Reading

- *Fundamentals of Classical Thermodynamics* (SI Version): Evaluation of Actual Combustion Processes. (Wylen and Sonntag 1986)
- *Thermal Engineering*: Capacity and Efficiency of Steam Generating Units. (Solberg, n.d.)
- *Basic Engineering Thermodynamics in SI Units*: Boiler calculations. (Joel 1996)

# Chapter 2

# Turbine Efficiency

Learning Objectives

Operate the Plant at the following generating capacities to compute the isentropic change in enthalpy and thermal efficiency for the HP turbine:

- 35% Load (I13),
- 80% Load (I14),
- 230 MW (I10).

## 2.1 Theory

Recall from the First and Second Law of Thermodynamics that the adiabatic process where entropy remains constant provides the maximum energy for work. As shown on the H-S coordinates, the difference in enthalpy, $(H_1 - H_2)$, is maximum when the lowest enthalpy $(H_2)$ is reached at the exit conditions. The ideal expansion is, therefore, a vertical line.

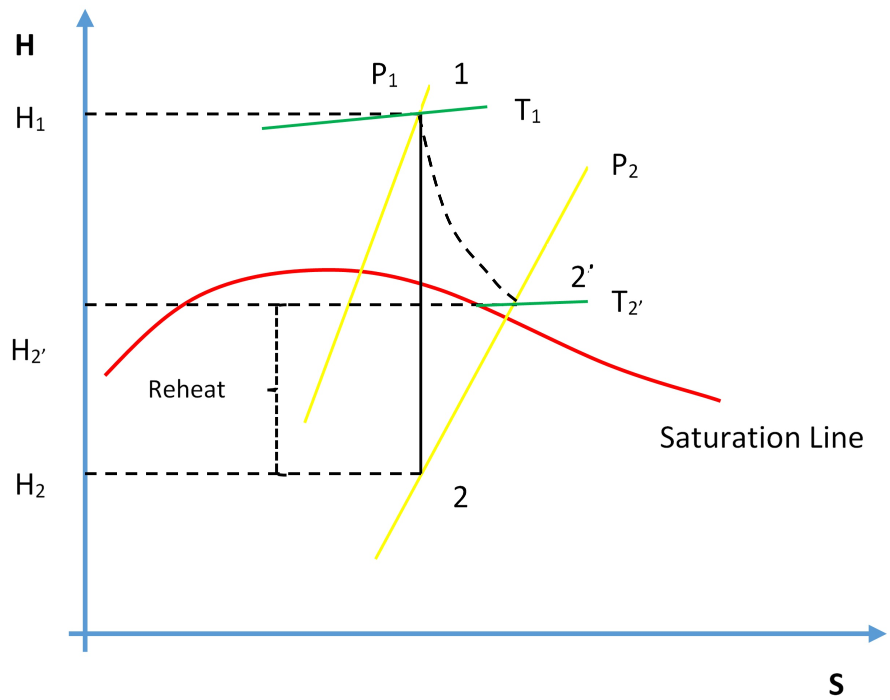



Figure 2.1: Turbine efficiency (H-S diagram).

On the diagram above, $T_1$, $P_1$, and $P_2$ are known process variables; for example, $H_1$ is determined by using $T_1$ and $P_1$. $H_2$ can then be found by drawing a vertical line from $P_1$ to $P_2$ by following adiabatic isentropic expansion (expansion at constant entropy).

Non-ideal processes, or real processes, however, do not present straight lines as shown on the Mollier diagram due to such factors as friction. If the expansion is not isentropic (i.e., entropy is not constant but increases), the lowest enthalpy $H_2$ cannot be reached at the exit conditions; in other words, $H_{2'} > H_2$. This means that $\Delta H$ for the ideal expansion is greater than $\Delta H$ for the non-ideal expansion between the same pressure boundaries. The internal turbine efficiency is therefore given by:

$$\eta_{\text{Turbine}} = \frac{\text{Actual change in enthalpy}}{\text{Isentropic change in enthalpy}}$$

$$\eta_{\text{Turbine}} = \frac{(H_1 - H_{2'})}{(H_1 - H_2)}$$

The difference in enthalpy $H_{2'} - H_2$ is called the reheat factor and is the basis for multi-stage turbines. As can be seen on the Mollier diagram, the pressure curves are divergent. This means that the higher the pressure drop in a single-stage turbine, the greater the reheat factor and in turn the lower the turbine efficiency. However, if the steam is expanded through multiple stages and between each stage the steam is reheated, higher turbine efficiencies can be achieved. We will see this effect later in the Power Plant Efficiency lab.

Lab Instructions

You will run 3 different initial conditions in this lab:
- 35% Load (I13),
- 80% Load (I14),
- 230 MW (I10).

For each condition, collect the relevant data to compute the isentropic change in enthalpy for the HP turbine. Compare your results — which of the three conditions yields the most favourable results and why?

## 2.2 Hints & Tips

In addition to various pressure and temperature values, log the following tags in your trends:

- Z03020
- E03018

To calculate the enthalpy values, you may use an app or online tool such as the Superheated Steam Table.

**!** Deliverables

Your lab report is to include the following:
- **Trend plots:** Supply all plots taken for each of the 3 conditions,
- **Computation:** Calculate the turbine efficiency for the 3 conditions specified,
- **Conclusion:** Write a summary (max. 500 words; use a text box if using Excel) comparing your results and suggestions for further study.

## 2.3 Further Reading

- *Thermodynamics and Heat Power*: Vapor power cycles. (Granet and Bluestein 2015)

# Chapter 3

# Power Plant Efficiency

ℹ Learning Objectives

Operate the Plant at full generating capacity and compute the Power Plant Efficiency when the plant is operating:

- Under normal conditions,
- With the cooling water temperature very high (lake water temperature: 35°C),
- Without regeneration.

## 3.1 Theory

Excluding hydroelectric power plants, most power generating plants employ a type of boiler and steam turbine. High-pressure steam leaves the boiler and enters the turbine. The steam expands in the turbine and does work which enables the turbine to drive the electric generator. The exhaust steam leaves the turbine and enters the condenser where heat is transferred from the steam to cooling water. The pressure of the condensate leaving the condenser is increased in the pump, thereby enabling the condensate to flow into the boiler. This thermodynamic cycle is known as the **Rankine Cycle**.

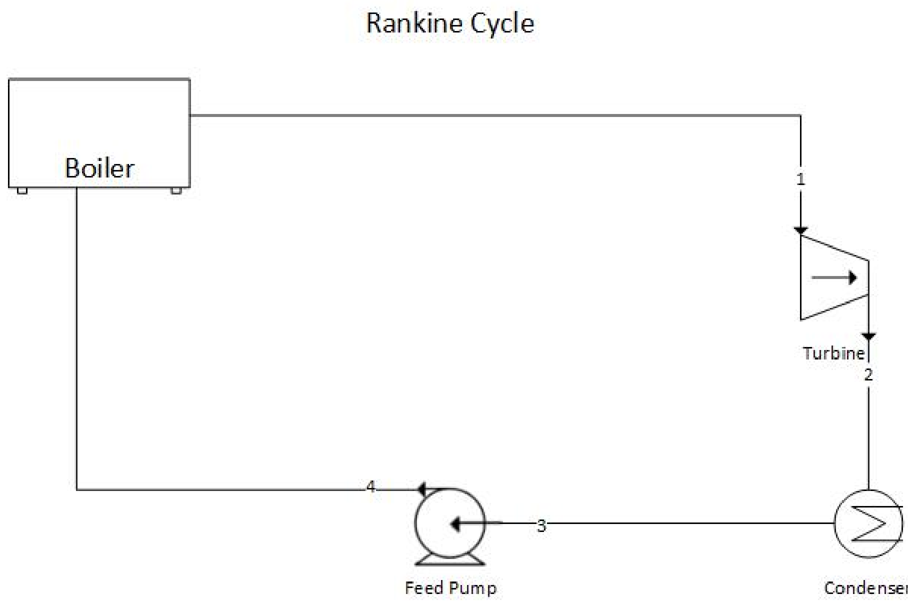



Figure 3.1: Schematic diagram of a steam power plant

## 3.2 The Rankine Cycle Efficiency

Some heat is always lost from the steam to cooling water. In addition, feed pumps consume energy, thus reducing the net work output. The Rankine Cycle efficiency is expressed as:

$$\eta_{\text{Rankine}} = \frac{\text{Net work output}}{\text{Heat supplied in the boiler}}$$

or equivalently:

$$\eta_{\text{Rankine}} = \frac{W_{\text{Turbine}} - W_{\text{Pump}}}{Q_{\text{boiler}}}$$

Referring to the schematic diagram and using the enthalpy values at each state point in the Rankine cycle:

$$\eta_{\text{Rankine}} = \frac{(h_1 - h_2) - (h_4 - h_3)}{h_1 - h_4}$$

## 3.3 Improvements to the Rankine Cycle Efficiency

### 3.3.1 Effect of Pressure and Temperature

If the exhaust pressure drops from $P_4$ to $P_4'$, with the corresponding decrease in the temperature at which heat is rejected in the condenser, the net work increases. Similarly, if the steam is superheated in the boiler, the work output is further increased. Superheating is achieved by increasing the time the steam is exposed to the flue gases. As a result, for a given power output, a plant using superheated steam will be smaller in size than one using dry saturated steam.

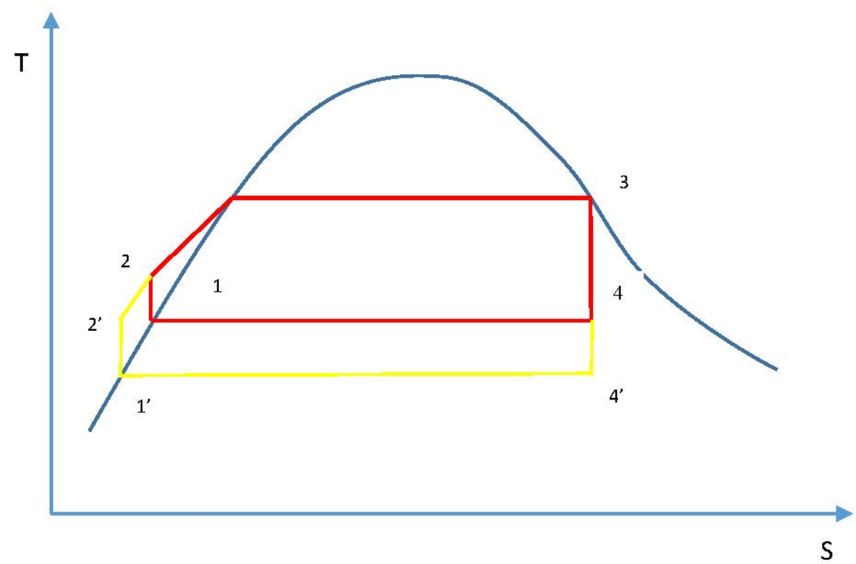



Figure 3.2: Effect of exhaust pressure

### 3.3.2 The Reheat Cycle

The efficiency of the Rankine cycle is increased by superheating the steam. To improve efficiency further without requiring exotic high-temperature materials, the **reheat cycle** has been developed. In this cycle, steam expands to some intermediate pressure in the high-pressure turbine and is then reheated in the boiler, after which it expands in the low-pressure turbine to the exhaust pressure.

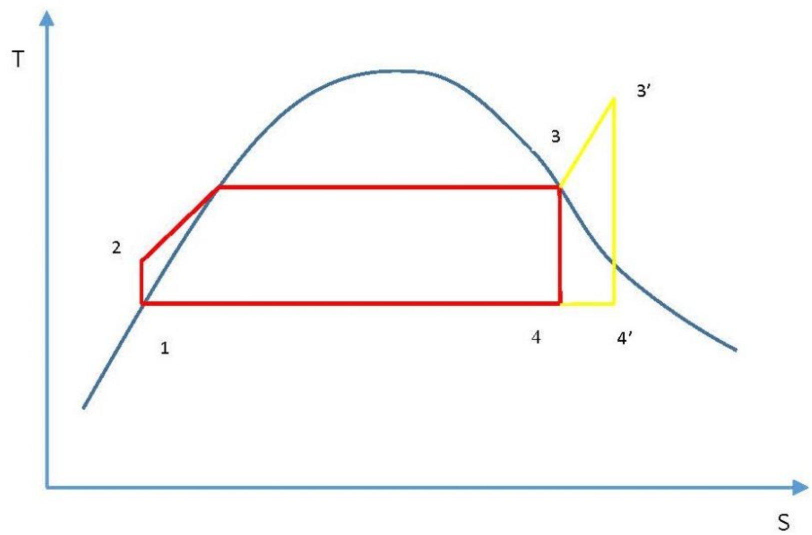



Figure 3.3: Effect of superheating

The thermal efficiency of the Rankine cycle with reheat is:

$$\eta_{\text{thermal}} = \frac{W_{12} + W_{67} - W_{43}}{Q_{41} + Q_{26}}$$

### 3.3.3 The Regenerative Cycle

Another variation of the Rankine cycle is the **regenerative cycle**, which uses feedwater heaters. In a simple Rankine cycle, feedwater heating from state 2 to 2' occurs at a much lower average temperature than the vaporisation process from 2' to 3. Consequently, the average temperature at which heat is supplied is lower than in the corresponding Carnot cycle, and the Rankine cycle efficiency is therefore less than that of the Carnot cycle.

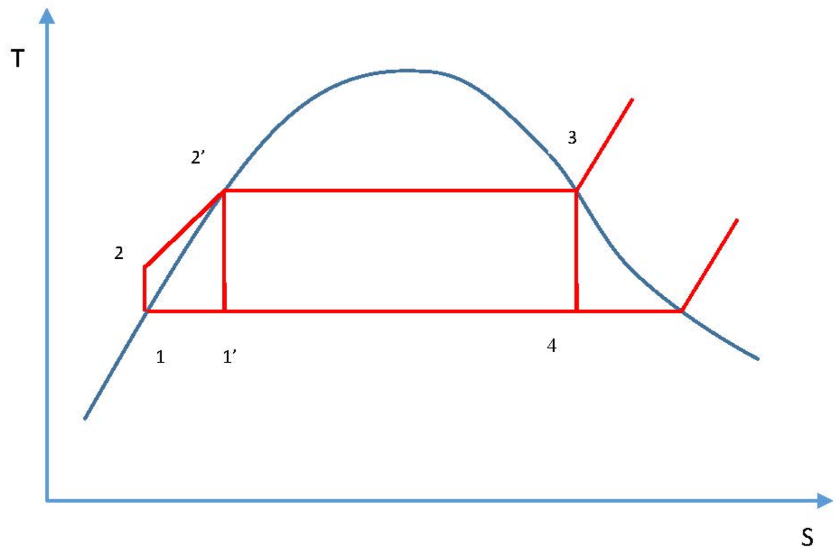



Figure 3.4: Relationship between Carnot cycle and Rankine cycle

In the regenerative cycle, feedwater enters the boiler at a point between states 2 and 2', raising the average temperature at which heat is supplied and thus improving efficiency.

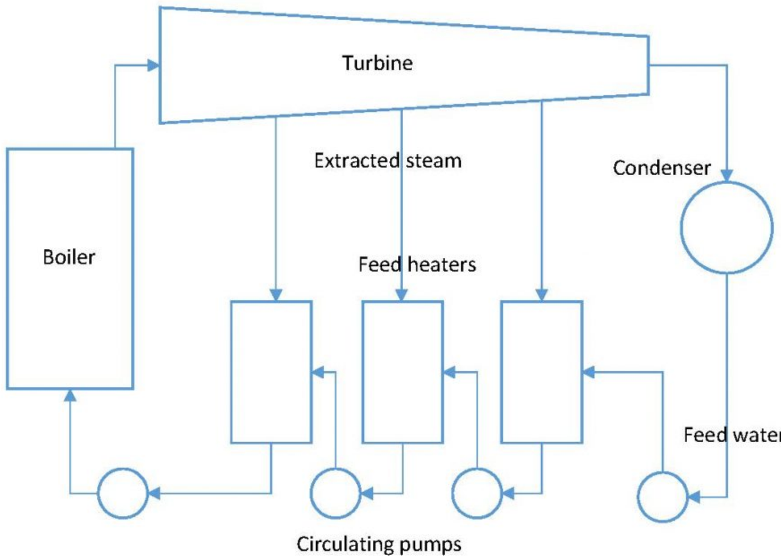



Figure 3.5: Regenerative cycle

## 3.4 Plant Thermal Efficiency

To calculate the overall plant thermal efficiency, the heat added in the reheater sections of the boiler must be included:

$$\eta_{\text{thermal}} = \frac{W_{\text{Turbines}} - W_{\text{Pumps}}}{Q_{\text{boiler}} + Q_{\text{Reheat1}} + Q_{\text{Reheat2}}}$$

Lab Instructions

Run the initial condition **I10 230 MW_oil_auto** and complete the following:

1. Draw a T–S diagram of the Rankine cycle (not to scale) including reheat and regeneration.
2. Using the Trend Group Directory, collect the relevant process values.
3. Calculate the overall thermal efficiency of the plant under three conditions:
   - **Normal conditions**,
   - **High cooling water temperature** — set Variable List Page 0100, tag `T00305` to 35°C,

- **No regeneration** — close all steam extraction valves and set `T00305` to 10°C.

## 3.5 Hints & Tips

The following tags must be logged in your trends:

| Tag | Description |
|---|---|
| `Q02395` | Reheater 1 transferred heat |
| `Q02375` | Reheater 2 transferred heat |

- For **Boiler Feedwater Inlet Temperature**, use the Startup Heat Exchanger Feedwater Outlet Temperature (tag `T02447`).
- For the third calculation, make sure you closed all steam extraction valves and set T00305 to 10°C:
- To calculate enthalpy values, use a steam tables app or an onlinesuperheated steam table.

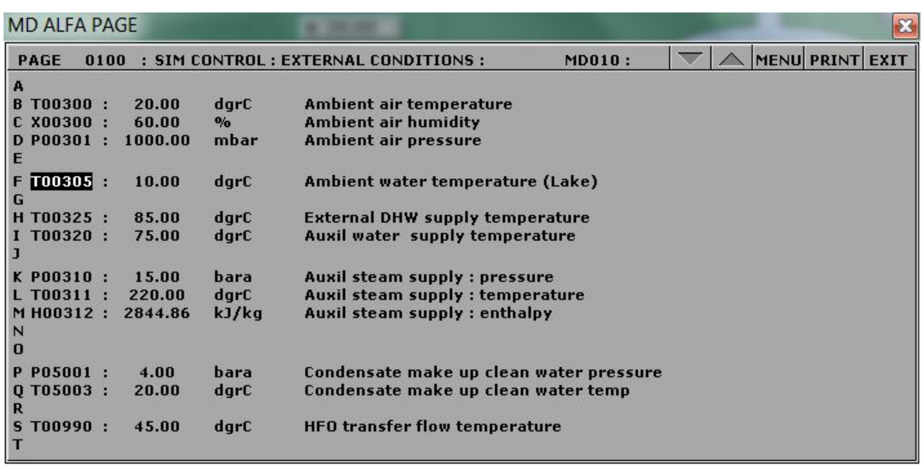



Figure 3.6: Lake water temperature setting

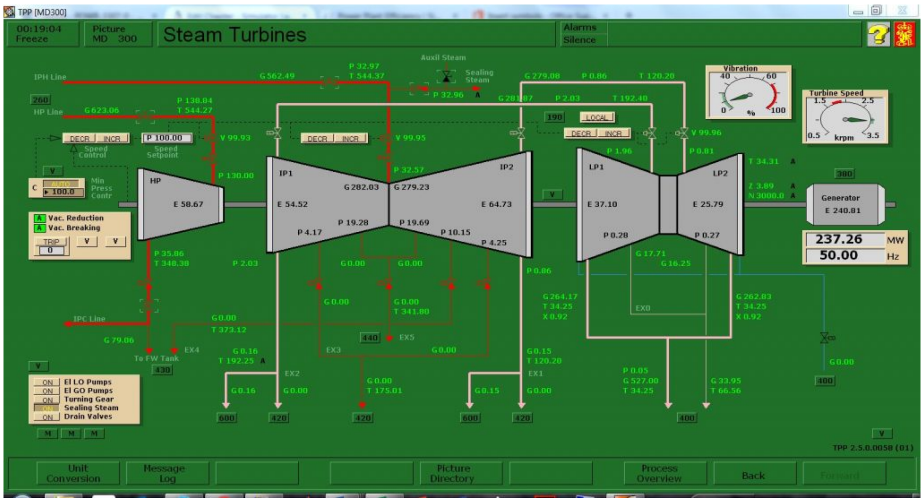



Figure 3.7: No steam extraction

**!** Deliverables

Your lab report must include the following:

1. **T–S Diagram** — As per the lab instructions above.
2. **Trend Plots** — Supply all plots taken during the lab.
3. **Computation** — Calculate the overall thermal efficiency of the plant under each of the three conditions.
4. **Conclusion** — A summary (maximum 500 words) comparing your results and suggesting areas for further study.

## 3.6 Further Reading

- *Applied Thermodynamics for Engineering Technologists* — Steam Plant. (Eastop and McConkey 1993)
- *Fundamentals of Classical Thermodynamics (SI Version)* — Vapour Power Cycles. (Wylen and Sonntag 1986)
- *Thermodynamics and Heat Power* — Vapour Power Cycles. (Granet and Bluestein 2015)

# Chapter 4

# $NO_x$ Emissions

**Learning Objectives**

Operate the Plant at full generating capacity and compare the $NO_x$ emissions when the plant is operating:

- Under normal conditions,
- During over-fire air damper failure,
- During over-burner air control failure,
- Burning poor quality fuel,
- With the $DeNO_x$ plant bypassed.

## 4.1 Theory

Combustion of coal generates considerable quantities of byproducts, some of which are considered pollutants. The byproducts are mostly water vapour, carbon dioxide, and nitrogen — all readily found in the atmosphere — and do not necessarily pose a direct health hazard. However, because of the large quantities of coal combusted, even small concentrations of pollutants in the flue gas translate into large absolute quantities of hazardous emissions. The main pollutants that can cause health problems are sulfur oxides, nitrogen oxides, particulate matter, and trace elements such as arsenic, lead, and mercury.

During combustion in a coal-fired power plant, nitrogen present in both the coal and the combustion air is converted into nitric oxide (NO) and nitrogen dioxide ($NO_2$). These nitrogen oxides are collectively referred to as $\mathbf{NO_x}$. $NO_x$ emissions contribute to the formation of acid rain.

$NO_x$ is primarily formed by two mechanisms:

- **Thermal $NO_x$** — formed at high flame temperatures; its formation rate increases exponentially with combustion temperature.
- **Fuel-bound $NO_x$** — dependent on the nitrogen content of the fuel.

The most effective strategies to minimize $NO_x$ formation are to reduce flame temperature, reduce excess oxygen, and/or burn fuels with a low nitrogen content.

## 4.2 The DeNO$_x$ Plant

The purpose of the DeNO$_x$ plant is to remove nitrogen oxides from the flue gases using **selective catalytic reduction (SCR)**. The reducing agent is ammonia gas. The plant comprises two SCR reactors and an ash silo; various dampers channel flue gases either into or around (bypassing) the SCR reactors.

### 4.2.1 Operation

The DeNO$_x$ plant is highly automated and controlled by Programmable Logic Control (PLC) sequences:

| Sequence | Function |
|---|---|
| S701 | Purge |
| S702 / S703 | Start / Stop Reactors |
| S704 / S705 | Heating of Reactors |
| S706 / S707 | Ammonia Injection |
| S708 / S709 | Product Handling |
| S710 / S711 | Soot Blowing |

### 4.2.2 Over-Fire Air (OFA)

To reduce $NO_x$, a quantity of additional air is supplied above all burner planes. This **over-fire air (OFA)** promotes a fuel-rich mixture in the high-temperature combustion zone at the burners (two-step combustion), thereby reducing $NO_x$ formation. Approximately 10% of the total combustion air is supplied as OFA. OFA controller failure (MD200 malfunction 0881) is modeled in the simulator.

### 4.2.3 Over-Burner Air (OBA)

To further reduce $NO_x$ generation, a portion of the secondary air is split from the main duct and directed into a third channel immediately above the burner. The damper controlling this stream is abbreviated **OBA**. OBA controller failure (MD180 malfunction 0780) is modeled in the simulator.

### 4.2.4 Fuel Quality

In the simulator, the chemical composition of the coal or other fuels can be specified. The five elemental components — C, H, S, O, and N — should preferably sum to 100%; however, this is not strictly required, as the values are always normalized to a 100% basis before use in further computations. Water content and inert matter (ash/slag) are specified separately. Water content varies considerably and has a significant impact on the preheating requirement for primary air.

The simulator computes the lower heating value (accounting for water and inert matter), the theoretical combustion air required, and the flue gas produced. Air and flue gas quantities are reported in ncm/kg (normal cubic metres per kilogram).

Fuel data can be modified using Variable List page 0111 on MD180.

Lab Instructions

Run the initial condition **I14 80% Coal** and set up trends for the following variables:

`G02197  X17821  G17107  X17106  D17104`

`T17103  C08444  G08444  G08443  C08400`

1. **Stable operation** — After 5 minutes of stable operation, freeze the simulator and print both trend windows. This serves as the reference point for all subsequent steps.
2. **OFA damper failure** — Switch to run mode and activate malfunction **0881** on MD200. After 5 minutes, freeze the simulator and print both trends. Deactivate the malfunction before proceeding.
3. **OBA control failure** — Switch to run mode and activate malfunction **0780** on MD180. After 5 minutes, freeze the simulator and print both trends. Deactivate the malfunction before proceeding.
4. **Burning poor quality fuel** — Switch to run mode and access Variable List page 0111 on MD180. Enter the following values, then wait 5 minutes before freezing the simulator and printing both trends.

| Tag | Value |
|---|---|
| X00820 | 70.40 |
| X00821 | 5.10 |
| X00822 | 1.10 |
| X00823 | 12.50 |
| X00824 | 1.60 |
| X00825 | 9.30 |

5. **DeNO$_x$ plant bypassed** — Switch to run mode and bypass SCR 1 and SCR 2 on MD710 and MD720 respectively. After 5 minutes, freeze the simulator and print both trends.

## 4.3 Hints & Tips

- Ensure all trend printouts are labeled clearly; unlabeled plots make data analysis very difficult.
- Tabulate your collected data in a table matching the format shown in the lab manual.

**Percentage deviation** is defined as:

$$\text{Percentage deviation} = \frac{\text{Current Value} - \text{Reference Value}}{\text{Reference Value}} \times 100\%$$

Tabulate the percentage deviation for each operating condition alongside the raw measured values.

**!** Deliverables

Your lab report must include the following:

1. **Trend plots** — All plots taken during the lab, clearly labeled for each operating condition.
2. **Computation** — Calculate the percentage deviation for each condition, and plot your results.
3. **Conclusion** — A summary (maximum 500 words) comparing your results across operating conditions and suggesting areas for further study.

## 4.4 Further Reading

- BCIT. *Thermal Power Plant Simulator Course Manual* — The DeNO$_x$ Plant.

# Chapter 5

# Heat Exchangers

> **Learning Objective**
>
> Operate the Plant at full generating capacity and study the effect of heat exchanger surface area using the Low Pressure Feed Heater 3.

## 5.1 Theory

A heat exchanger is equipment in which heat exchange takes place between two working media that enter and exit at different temperatures. Its main function is either to remove heat from a hot working medium or to add heat to a cold working medium. Depending on the direction of fluid flow, a heat exchanger is classified as either a **parallel (concurrent) flow** or a **counter flow** heat exchanger.

In a **parallel flow** heat exchanger, both fluids enter at the same end, so their largest temperature difference occurs at the inlet. The temperature difference decreases progressively along the length of the exchanger. In a **counter flow** heat exchanger, fluids enter at opposite ends, so the temperature difference between the two fluids remains relatively constant along the length of the exchanger.

## 5.2 Heat Transfer Equations

The heat transferred by the hot and cold streams is given by:

$$Q_{\text{hot}} = \dot{m}_{\text{hot}}\, c_{p,\text{hot}}\, \Delta T_{\text{hot}}$$

$$Q_{\text{cold}} = \dot{m}_{\text{cold}}\, c_{p,\text{cold}}\, \Delta T_{\text{cold}}$$

Newton's Law of Cooling states that the rate of heat loss is proportional to the temperature difference between the body and its surroundings:

$$Q = \alpha \cdot A \cdot \Delta T$$

where $\alpha$ is the heat transfer coefficient [$\mathrm{W/m^2 \cdot K}$], $A$ is the surface area [$\mathrm{m^2}$], and $\Delta T$ is the temperature difference [K].

More generally, the heat transferred between the hot and cold streams can be written as:

$$Q = \frac{F \cdot \Delta T_{\text{LMTD}}}{R_T}$$

or equivalently:

$$Q = F \cdot (UA) \cdot \Delta T_{\text{LMTD}}$$

where $F$ is the correction factor (equal to 1 for this SIMLAB; typically between 0.5 and 1), $R_T$ is the overall thermal resistance, $U$ is the overall heat transfer coefficient, and $\Delta T_{\text{LMTD}}$ is the Log Mean Temperature Difference.

## 5.3 Overall Thermal Resistance

The overall thermal resistance is the sum of three resistances in series:

$$R_T = R_{hf} + R_w + R_{cf}$$

$$R_{hf} = \frac{1}{A_1\, \alpha_h}$$

$$R_w = \frac{\ln\left(\dfrac{D_2}{D_1}\right)}{2\pi L\, \lambda_w}$$

$$R_{cf} = \frac{1}{A_2\, \alpha_c}$$

## 5.4 Heat Transfer Coefficients

The local heat transfer coefficients $\alpha_h$ and $\alpha_c$ are obtained from the Nusselt number correlations below.

**For the hot (cooling) side:**

$$\alpha_h = \frac{Nu_h \, \lambda_h}{D_h}, \qquad Nu_h = 0.3 \, Re_h^{0.8} \, Pr_h^{0.3}$$

**For the cold (heating) side:**

$$\alpha_c = \frac{Nu_c \, \lambda_c}{D_c}, \qquad Nu_c = 0.3 \, Re_c^{0.8} \, Pr_c^{0.3}$$

## 5.5 Log Mean Temperature Difference

The LMTD is defined in terms of the temperature differences at each end (subscripts 1 and 2) of the heat exchanger:

$$\Delta T_{\mathrm{LMTD}} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\dfrac{\Delta T_1}{\Delta T_2}\right)}$$

## 5.6 Heat Exchanger Effectiveness

Effectiveness differs from efficiency: efficiency concerns minimising waste, whereas effectiveness concerns maximising output. Heat exchanger effectiveness is defined as the ratio of the actual heat transfer rate to the maximum possible heat transfer rate for the given inlet temperatures:

$$\varepsilon = \frac{Q}{Q_{\max}}$$

$$Q_{\max} = C_{\min} \, (T_{h,i} - T_{c,i})$$

where $C_{\min}$ is the smaller of the two fluid heat capacity rates:

$$C_h = \dot{m}_h \, c_{p,h}, \qquad C_c = \dot{m}_c \, c_{p,c}$$

Lab Instructions

Run the initial condition **I10 230 MW_oil_auto** and set up trends for the following variables:

| Variable | Tag |
|---|---|
| Heat Area Factor | C34201 |
| $Q$ | Q34228 |
| $T_{\text{cold},1}$ | T24214 |
| $T_{\text{cold},2}$ | T34214 |
| $\dot{m}_{\text{cold}}$ | G34213 |
| $T_{\text{hot},1}$ | T34204 |
| $T_{\text{hot},2}$ | T34227 |
| $\dot{m}_{\text{hot}}$ | G34203 |

1. **Heat Area Factor = 0.5** — Using MD420 and Variable List 4210, set `C34201` to **0.5**. Run the simulator for 15 minutes, then freeze and print both trends.
2. **Heat Area Factor = 1.0** — Set `C34201` to **1.0** (the default value for the heat transfer coefficient × area product). Run the simulator for 15 minutes, then freeze and print both trends.
3. **Heat Area Factor = 1.5** — Set `C34201` to **1.5**. Run the simulator for 15 minutes, then freeze and print both trends.

## 5.7 Hints & Tips

- Label all trend printouts with descriptive names; unlabeled plots make comparison very difficult.
- You are varying the Heat Area Factor ($U \cdot A$) of LP Feed Heater 3 across three levels and comparing the resulting temperature profiles.
- Your evaluation will be based on the computed **LMTD** and **Heat Exchanger Effectiveness** ($\varepsilon$) values for each test.

Deliverables

Your lab report must include the following:

1. **Trend plots** — All plots taken during the lab, clearly labelled for each Heat Area Factor setting.
2. **Computation** — Use MATLAB or MS Excel to calculate the LMTD and heat exchanger effectiveness for each of the three tests.
3. **Conclusion** — A summary (maximum 500 words) comparing your results across the three settings and suggesting areas for further

study.

## 5.8 Further Reading

- *Applied Thermodynamics for Engineering Technologists* — Heat Transfer. (Eastop and McConkey 1993)

# Chapter 6

# Combustion Analysis

**Learning Objectives**

Operate the Plant at 80% capacity burning coal to:

- Perform combustion analyses for two types of coal,
- Compare results.

## 6.1 Theory

In Chapter 1, combustion efficiency was defined as the ratio of the burner's capability to burn fuel completely to the unburned fuel and excess air in the exhaust. In this lab, we carry out a full combustion analysis.

Fossil fuels may be classified as solid, liquid, or gaseous. The vast majority are based on carbon (C), hydrogen ($H_2$), or some combination of the two called hydrocarbons. During combustion, oxygen ($O_2$) combines rapidly with C, $H_2$, sulphur (S), and their compounds, releasing energy. Except for special high-temperature applications (e.g. oxyacetylene welding), the $O_2$ required for combustion is obtained from air.

For engineering purposes, air is considered to have the following composition by mass:

| Component | Mass fraction |
|---|---|
| $O_2$ | 23% |
| $N_2$ | 77% |

The proportions in which elements enter the combustion reaction depend on their relative molecular weights:

| Element | Symbol | Molecular Weight |
|---|---|---|
| Carbon | C | 12 |
| Sulphur | S | 32 |
| Hydrogen | $H_2$ | 2 |
| Oxygen | $O_2$ | 32 |
| Nitrogen | $N_2$ | 28 |

## 6.2 Stoichiometric Combustion Theory

Complete combustion of a simple hydrocarbon fuel produces carbon dioxide ($CO_2$) from the carbon and water ($H_2O$) from the hydrogen. For a hydrocarbon fuel of general composition $C_nH_m$, the combustion equation on a molar basis is:

$$\mathrm{C}_n\mathrm{H}_m + y\,\mathrm{O}_2 \rightarrow a\,\mathrm{CO}_2 + b\,\mathrm{H}_2\mathrm{O}$$

The element balances are:

**Carbon balance:**

$$a = n \quad [\text{kmol CO}_2/\text{kmol fuel}]$$

**Hydrogen balance:**

$$2b = m \implies b = \frac{m}{2} \quad [\text{kmol H}_2\text{O}/\text{kmol fuel}]$$

**Oxygen balance:**

$$2y = 2a + b \implies y = a + \frac{b}{2} \quad [\text{kmol O}_2/\text{kmol fuel}]$$

In practice, combustion occurs in air rather than pure oxygen. Nitrogen in the air may react to produce nitrogen oxides, and fuels often contain elements other than carbon and hydrogen. Furthermore, combustion is rarely complete — exhaust gases may contain unburned or partially burned products in addition to $CO_2$ and $H_2O$.

For further calculations, air is treated as a mixture of 21 mol% $O_2$ and 79 mol% $N_2$, with nitrogen acting as an inert gas. The stoichiometric combustion equation for $C_nH_m$ in air then becomes:

$$\mathrm{C}_n\mathrm{H}_m + y\left(\mathrm{O}_2 + \frac{79}{21}\,\mathrm{N}_2\right) \rightarrow a\,\mathrm{CO}_2 + b\,\mathrm{H}_2\mathrm{O} + c\,\mathrm{N}_2$$

with the additional nitrogen balance:

**Nitrogen balance:**

$$c = y\,\frac{79}{21} \quad [\text{kmol N}_2/\text{kmol fuel}]$$

Flue gas compositions are expressed as mole fractions in kmol of product per kmol of fuel.

## 6.3 Worked Example: Combustion of Octane in Air

Determine the stoichiometric air/fuel mass ratio and product gas composition for combustion of octane ($C_8H_{18}$) in air.

$$C_8H_{18} + y\left(O_2 + \frac{79}{21}\,N_2\right) \rightarrow a\,CO_2 + b\,H_2O + c\,N_2$$

**Carbon balance:**

$$a = 8 \quad [\text{kmol CO}_2/\text{kmol fuel}]$$

**Hydrogen balance:**

$$b = \frac{18}{2} = 9 \quad [\text{kmol H}_2\text{O}/\text{kmol fuel}]$$

**Oxygen balance:**

$$y = 8 + \frac{9}{2} = 12.5 \quad [\text{kmol O}_2/\text{kmol fuel}]$$

**Nitrogen balance:**

$$c = 12.5 \times \frac{79}{21} = 47 \quad [\text{kmol N}_2/\text{kmol fuel}]$$

The balanced combustion equation is:

$$C_8H_{18} + 12.5\left(O_2 + \frac{79}{21}\,N_2\right) \rightarrow 8\,CO_2 + 9\,H_2O + 47\,N_2$$

**Air/fuel mass ratio** — the molar mass of octane is 8(12) + 18(1) = 114 kg/kmol:

$$\frac{12.5\text{ kmol O}_2}{\text{kmol fuel}} \times \frac{1\text{ kmol fuel}}{114\text{ kg fuel}} \times \frac{32\text{ kg O}_2}{\text{kmol O}_2} \times \frac{100\text{ kg air}}{23\text{ kg O}_2} = 15.25\,\frac{\text{kg air}}{\text{kg fuel}}$$

**Flue gas composition (molar basis):**

Total kmol of flue gas $= 8 + 9 + 47 = 64$ kmol/kmol fuel

| Species | Mole fraction |
|---|---|
| $CO_2$ | $8/64 = 12.5\%$ |
| $H_2O$ | $9/64 = 14.1\%$ |
| $N_2$ | $47/64 = 73.4\%$ |

Other fuel impurities complicate the calculation. Sulphur, for instance, typically combusts to sulphur dioxide ($SO_2$). Ash — the noncombustible inorganic mineral content of the fuel — is assumed inert and is neglected in the calculations below.

## 6.4 Mass-Based Combustion Chemistry

For solid and liquid fuels, composition is usually reported on a mass basis from an ultimate analysis. The stoichiometric combustion reactions, written using molecular weights, are:

**Carbon:**

$$\mathrm{C} + \mathrm{O_2} \rightarrow \mathrm{CO_2} \quad \Longrightarrow \quad 12 + 32 = 44 \quad \Longrightarrow \quad 1 \text{ kg C} + 2\tfrac{2}{3} \text{ kg } \mathrm{O_2} = 3\tfrac{2}{3} \text{ kg } \mathrm{CO_2}$$

**Hydrogen:**

$$2\mathrm{H_2} + \mathrm{O_2} \rightarrow 2\mathrm{H_2O} \quad \Longrightarrow \quad 4 + 32 = 36 \quad \Longrightarrow \quad 1 \text{ kg } \mathrm{H_2} + 8 \text{ kg } \mathrm{O_2} = 9 \text{ kg } \mathrm{H_2O}$$

**Sulphur:**

$$\mathrm{S} + \mathrm{O_2} \rightarrow \mathrm{SO_2} \quad \Longrightarrow \quad 32 + 32 = 64 \quad \Longrightarrow \quad 1 \text{ kg S} + 1 \text{ kg } \mathrm{O_2} = 2 \text{ kg } \mathrm{SO_2}$$

Since air contains 23.2% $O_2$ by mass, 1 kg of $O_2$ is contained in $1/0.232 = 4.3$ kg of air (the **stoichiometric mass of air**), which in turn contains $4.3 - 1 = 3.3$ kg of $N_2$. For sulphur combustion in air:

$$1 \text{ kg S} + 4.3 \text{ kg air} \rightarrow 2 \text{ kg } \mathrm{SO_2} + 3.3 \text{ kg } \mathrm{N_2}$$

## 6.5 Procedure

When fuel composition is given on a mass basis, follow these steps:

1. **Total $O_2$ required** — Determine the mass of $O_2$ required to combust each constituent and sum the results. Subtract any $O_2$ already present in the fuel.

2. **Stoichiometric air:**

$$\dot{m}_{\text{air, stoich}} = \frac{\dot{m}_{O_2\text{, required}}}{0.232}$$

3. **Total mass of combustion products** — Calculate the mass of each product. For example, for a carbon content of 84.9%:

$$m_{CO_2} = \frac{84.9}{100} \times 3\tfrac{2}{3} = 3.11 \text{ kg } CO_2/\text{kg fuel}$$

   Sum all product masses to obtain the total.

4. **Combustion product analysis by mass** — Express each product as a percentage of the total flue gas mass. For example, if the total flue gas mass is 12.09 kg/kg fuel:

$$x_{CO_2} = \frac{3.11}{12.09} \times 100 = 25.7\%$$

   Repeat for each constituent ($H_2O$, $SO_2$, $N_2$).

Lab Instructions

Run the initial condition **I14 80% Coal** and set up trends for the following variables:

`X00820 X00821 X00822 X00823 X00824 E23356`

`G02196 G02197 X32419 X02419 G00831 H00830`

1. **Burning default coal** — After 10 minutes of stable operation, freeze the simulator and print both trends. This serves as the reference point for the next step.
2. **Burning poor quality coal** — Switch to run mode and access Variable List page 0111 on MD180. Enter the values below, then run for 10 minutes before freezing and printing both trends.

| Tag | Value |
|---|---|
| X00820 | 70.40 |
| X00821 | 5.10 |
| X00822 | 1.10 |
| X00823 | 12.50 |
| X00824 | 1.60 |
| X00825 | 9.30 |

3. **Computation** — For both fuel types, calculate:
   - Total $O_2$ required,
   - Stoichiometric air,
   - Total mass of combustion products,
   - Mass percentage of each combustion product: $CO_2$, $H_2O$, $SO_2$, $N_2$.
4. **Comparison** — Compare your findings for the two fuels using the following simulator outputs:
   - Furnace outlet $SO_x$ flow,
   - Furnace outlet $NO_x$ flow,
   - CO content in flue gas,
   - $O_2$ content in flue gas,
   - Theoretical combustion air,
   - Coal heat value.

## 6.6 Hints & Tips

- Label all trend printouts clearly for each fuel; unlabeled plots make comparison very difficult.
- To change the fuel composition, use Variable List page 0111 on MD180.

**!** Deliverables

Your lab report must include the following:

1. **Trend plots** — All plots taken for each of the two fuels, clearly labeled.
2. **Computation** — Combustion analyses for both fuels performed in MATLAB or MS Excel, as per the procedure above.
3. **Conclusion** — A summary (maximum 500 words) comparing your results for the two fuel types and suggesting areas for further study.

## 6.7 Further Reading

- *Basic Engineering Thermodynamics in SI Units* — Combustion. (Joel 1996)
- *Thermal Engineering* — Fossil Fuels and Their Combustion. (Solberg, n.d.)
- Wikipedia. Stoichiometry.

# References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